SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I ***Patan Reddy Suhail (192312671L)*** *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
Suhail

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

Ans : **Given**

- File size = **150 MB**
- Bandwidth = **50 Mbps**
- Segment size = **1500 bytes**
- Data Link overhead = **40 bytes/frame**

Step 1: File size in bytes

$$150 \times 10^6 = 150{,}000{,}000 \text{ bytes}$$

Step 2: Number of segments

$$\frac{150{,}000{,}000}{1500} = 100{,}000 \text{ segments}$$

Each segment becomes one frame.

Frames = 100,000

Step 3: Total Data Link overhead

$$100{,}000 \times 40 = 4{,}000{,}000 \text{ bytes} = 4 \text{ MB}$$

Step 4: Total transmitted data

$$150 + 4 = 154 \text{ MB}$$

Step 5: Transmission time

Convert to bits:

$$154 \times 8 = 1232 \text{ Mb} \frac{1232}{50} = 24.64 \text{ seconds}$$

Answer

- Segments = **100,000**
- Frames = **100,000**
- Overhead = **4 MB**
- Transmission time = **24.64 s**

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Ans :

Given

- Window size = 6
- Total frames = 48

Number of windows

$$\frac{48}{6} = 8$$

Retransmissions

One frame lost per window.

$$8 \times 1 = 8$$

Answer

- Windows = **8**
- Retransmissions = **8**

Flow Control

Flow control prevents the sender from overwhelming the receiver, improving efficiency and avoiding buffer overflow.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

Ans: **Given**

- Packet = 12,000 bytes
- MTU = 1500 bytes
- Header = 20 bytes

Maximum payload per fragment

$$1500 - 20 = 1480 \text{ bytes}$$

Fragments

$$\frac{12000}{1480} = 8.1$$

Hence

$$9 \text{ fragments}$$

Header overhead

$$9 \times 20 = 180 \text{ bytes}$$

Answer

- Fragments = **9**
- Header overhead = **180 bytes**

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Ans:

Given

- Window size = 6
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Answer

- Windows = **8**
- Retransmissions = **8**

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

Ans:

Given

- File = 600 MB
- Checkpoint every 60 MB
- Failure after 390 MB

Checkpoints

At

60,120,180,240,300,360

Total

6 checkpoints

Data retransmitted

Last checkpoint = 360 MB

Failure = 390 MB

$$390 - 360 = 30 \text{ MB}$$

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

Ans:

Given

- Original = 900 MB
- Compression = 40%
- Encryption overhead = 5%

Compressed size

$$900 \times 0.60 = 540 \text{ MB}$$

Encrypted size

$$540 \times 1.05 = 567 \text{ MB}$$

Reduction

$$\frac{900 - 567}{900} \times 100 = 37\%$$

Answer

- Compressed size = **540 MB**
- Encrypted size = **567 MB**
- Reduction = **37%**

7.An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

Ans: **Given**

- File = 3 GB = 3000 MB
- Throughput = 120 Mbps
- Request size = 3 MB

Requests

$$\frac{3000}{3} = 1000$$

Transmission time

Convert file into megabits

$$3000 \times 8 = 24000 \text{ Mb} \frac{24000}{120} = 200 \text{ seconds}$$

Answer

- Requests = **1000**
- Time = **200 seconds**

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

Ans : Processing delay

$$4 + 3 + 5 + 2 + 1 = 15 \text{ ms}$$

Total delay

$$15 + 18 + 12 = 45 \text{ ms}$$

Answer

- **Total end-to-end delay = 45 ms**

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

Ans :

Given

- Useful data = 80 MB
- Frame size = 1600 bytes
- Overhead = 80 bytes

Useful payload/frame

$$1600 - 80 = 1520 \text{ bytes}$$

Frames

$$\frac{80{,}000{,}000}{1520} = 52{,}632 \text{ frames}$$

Total overhead

$$52{,}632 \times 80 = 4{,}210{,}560 \text{ bytes}$$

≈ **4.21 MB**

Efficiency

$$\frac{1520}{1600} \times 100 = 95\%$$

Answer

- Frames = **52,632**

- Overhead = **4,210,560 bytes (≈4.21 MB)**
- Efficiency = **95%**

10.A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

Ans :

Given

- Original = 240 MB
- Compression = 30%
- Segment size = 1400 bytes
- Data Link overhead = 60 bytes

Compressed size

$$240 \times 0.70 = 168 \text{ MB}$$

Convert to bytes

$$168{,}000{,}000 \text{ bytes}$$

Segments

$$\frac{168{,}000{,}000}{1400} = 120{,}000$$

Protocol overhead

$$120{,}000 \times 60 = 7{,}200{,}000 \text{ bytes} = 7.2 \text{ MB}$$

Final transmitted data

$$168 + 7.2 = 175.2 \text{ MB}$$

Answer

- Compressed size = **168 MB**
- Segments = **120,000**
- Protocol overhead = **7.2 MB**
- Final transmitted data = **175.2 MB**